03/10/2026 ITA0307 - Mobile Computing P. Sudhaithay Reddy
192372204
CO1 - AT2
CSE - A1

① 45MHz = 45,000 KHz.

full duplex needs $2 \times 25 = 50$KHz

Channels = $45,000 / 50 = 900$

∴ 900 full-duplex channels

②

channels/cell = $560/7 = 80$

∴ 80 channels/cell

③

Total Voice Channels = $100 \times 8 = 800$

∴ 800 simultaneous channels

④

$10 \log_{10}(80/4) = 10 \log_{10}(20) = 13.01$dB

∴ 13.01dB

⑤ $C/I = -60 - (-92) = 32$ dB

⑥ Total Channels = $42000/300 = 140$

per cell = $140/7 = 20$

∴ 20 Channels/cell

⑦ Doubling distance gives $64/2^4 = 64/16 = 4$W

⑧ distance doubles $P \propto d^2 = 2^2 = 4$

⑨ $4 \times 15 = 60$ Channels

(10) $v = 100/3.6 = 27.78$ m/s

$f_D = vf/c = (27.78 \times 900 \times 10^3)/(3 \times 10^3) = 83.3$ Hz

(11) $5/8 = 0.625$ ml

(12) $24{,}000/200 = 120$

∴ 120 Carriers

(13) .8W = 8000mW ; $10 \log_{10}(8000) = 39.03$ dBm

Received = 39.03 − 33 = 6.03 dBm

(14) $FSPL(dB) = 32.44 + 20 \log_{10}(f\text{MHz}) + 20 \log_{10}(d\text{km})$

$= 32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5) = 105.98$ dB

(15) 6 × 250 = 15,000 Calls

(16) 3min = 180s

Active Call = 4000 × 180 = 720,000

(17) 150 × 25 kHz = 3750 kHz = 3.5 MHz

(18) 30 bands Create 29 gaps between them

29 × 40 kHz = 1160 kHz = 1.16 MHz

(19) Diameter = 2 × 1.5 = 3 km

Handoff / hour = 90/3 = 30

∴ 30 handoff / hour.

(20) $8 \times 40 \times 250 = 80,000$ users

(21) $R = 0.4$ km. Hexagonal area $= (3\sqrt{3}/2) R^2 = 0.4154$ km^2

Cells $= 400 / 0.4154 = 962.25$

$\therefore$ 963 cells

(22) $60,000 / 300 = 200$ channels

(23) Traffic $= 1200 \times 4 \times (2.5/60) = 200$ Erlangs

(24) $40,000 / 400 = 100$ users

(25) $1200 \times 600 = 720,000$ users

(26) Rounds $= 200/4 = 50$ rounds

(27) $40,000 \times 3 = 1,20,000$ Capacity $= 1,00,000$ B $= 20,000$

Fraction $= 20,000 / 120,000 \times 100 = 16.67\%$.

(28) $600,000 \times 0.16 = 96,000$

(29) $12,000 - 9,000 = 3000$ calls/s

(30) Total channels $= 48,000/200 = 240$

per cell $= 240/4 = 60$ channels

(31) $30,000/200 = 150$ channels

(32) $2500 \times 0.99 = 2475$ successful

(33) $12w = 12000mw$

$10 \log_{10}(12,000) = 40.79 dBm$

Received $= 40.79 - 45 = -4.21 dBm$

(34) $7 \times 900 = 6,300$ users

(35) $4.8/8 = 0.6 mJ$

(36) $v = 150/3.6 = 41.67 m/s$

$fD = vf/c = (41.67 \times 1.8 \times 10^3) / 3 \times 10^3 = 25 Hz$

(37) $70000 \times 4 = 280,000$

$B = 280,000 - 250,000 = 30,000$

B share $= 30,000/280,000 \times 100 = 10.71\%$

(38) $9 \times 600 = 5,400$ users

(39) $60,000 \times 0.004 = 240$ Call/h.

(40) path loss $= 43 - (-87) = 130 dB$

(41) $150 \times 2.5 = 375 Km^2$

(42) Channel $= 48,000/200 = 240$

per Cell $= 240/6 = 40$ channel